Code in Place 2026

Stanford CS106A

Section - Week 1

Welcome to Section!

David Tsai, Code in Place

Today’s Agenda

Introductions
Icebreakers

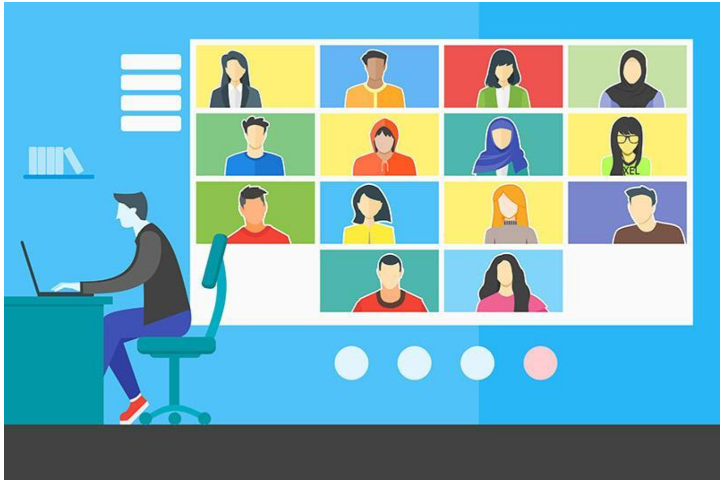
Norms
Guidelines

Concept Review
Karel, Control Flow

Practice Problems
“Hospital Karel”

Please Turn On Your Camera

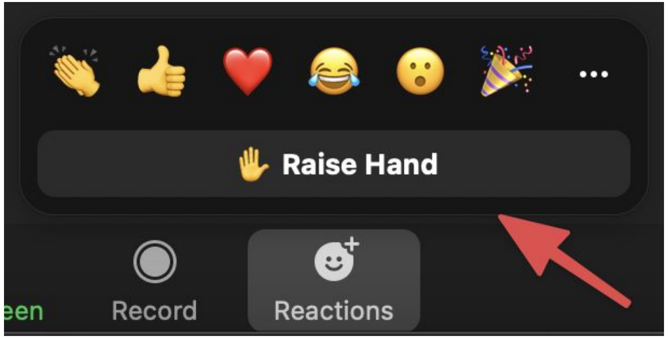
If you're able, please **turn on your camera**!
. . . . It can really make the section come to life!

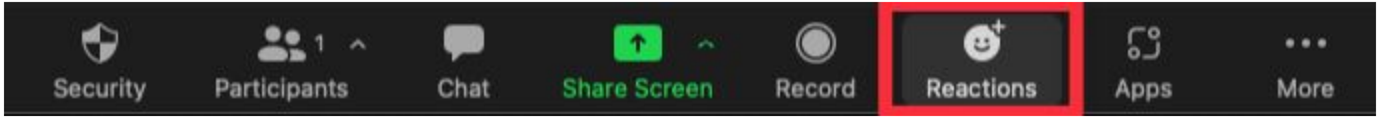


(Image source: https://as.virginia.edu/eight-ways-get-more-out-zoom)

Zoom Reactions

- 👍 **Thumbs Up:** If you understand.
- ✋ **Raise Hand:** If you have a question (or just speak in the mic).





Introductions

Hi, I'm David!

- Head TA, here at my 5th Code in Place.
    - Started as a CIP student! 2x volunteer Section Leader.
- CS @ Massachusetts Institute of Technology (MIT)
- Product Manager and former Software Engineer
- Love photography, video games, movies
- Guilty pleasure: Reality competition shows like Survivor
- Cool superpower I'd like to have:
    - Felix Felicis ("Liquid Luck" potion) from Harry Potter

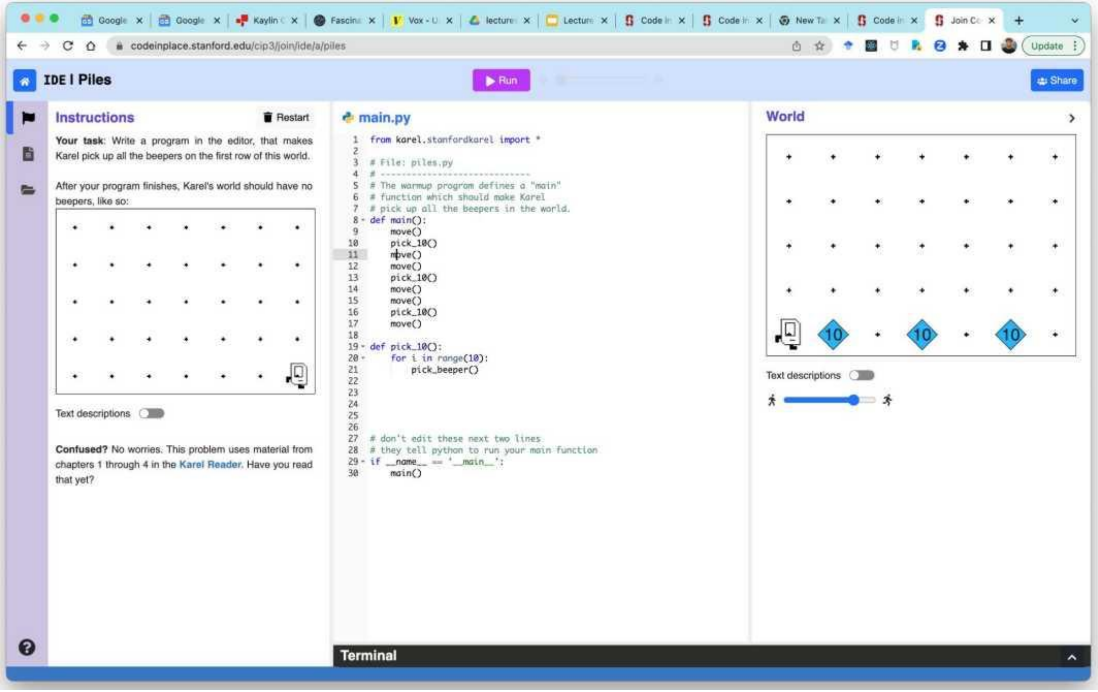
How about you?

- Where are you from?
- Your coding experience
- Cool superpower you'd like to have

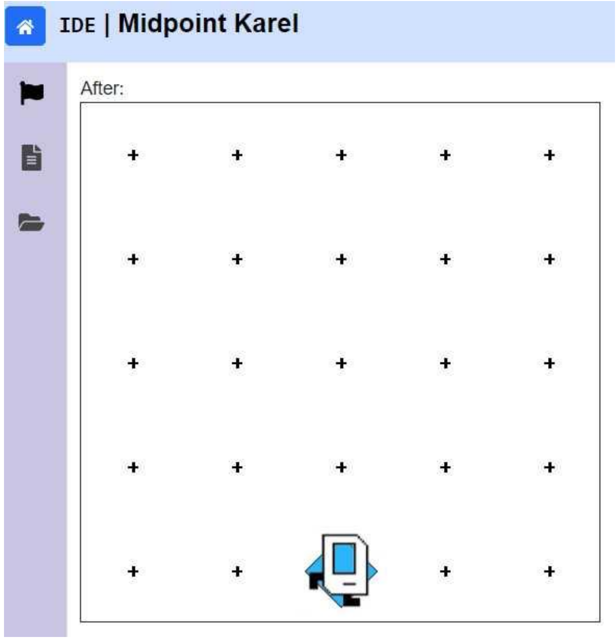
Course Weekly Timeline

Lectures	Section	Assignment
2 Lessons	**Live Zoom Session**	**Coding Challenges**
Both released every Monday. ~1 hour per lecture.	Practice lecture concepts. Prepare you for assignment.	Solidify your understanding. Optional: “Extension” projects

Brief Tour of Course’s Web Platform



Some Basic Python Functionality is TURNED OFF....for now!



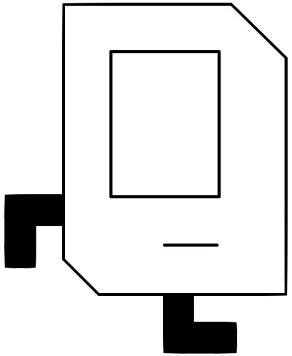
For the duration of Karel unit (Weeks 1 & 2):

- Basic Python functionality is turned off for simplicity
  - Focus on fundamentals of programming

- Examples of disabled functionality:
  - print statements
  - variables, lists, dictionaries
  - conditional keywords (`NOT`, etc.)
  - function return statements

Concepts Review (Lightning Round!)

Karel is like a LEGO set with only 4 types of blocks....but you can do a lot!



Karel only knows four commands at the beginning:

```
move()

turn_left()

put_beeper()

pick_beeper()
```

Helpful website to visualize Karel (by Sophia Westwood):

www.karelhelper.com

Control Flow

`for` **Loops**

`while` **Loops**

`if/else`

for **Loop**

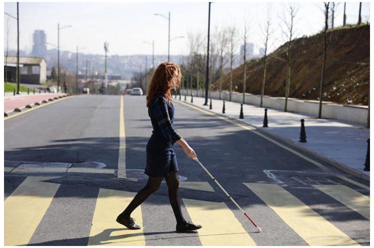
```
for i in range(count):
    statements                  # note indenting
```

```
# Example:
def turn_right():
    for i in range(3):
        turn_left()             # note indenting
```

`while` Loop

```
while condition:
    statements                # note indenting
```

```
# Example:
def move_to_wall():
    while front_is_clear():
        move()                # note indenting
```



Conditions Karel Can Check For

Test	*Opposite*	*What it checks*
`front_is_clear()`	`front_is_blocked()`	Is there a wall in front of Karel?
`left_is_clear()`	`left_is_blocked()`	Is there a wall to Karel's left?
`right_is_clear()`	`right_is_blocked()`	Is there a wall to Karel's right?
`beepers_present()`	`no_beepers_present()`	Are there beepers on this corner?
`beepers_in_bag()`	`no_beepers_in_bag()`	Any there beepers in Karel's bag?
`facing_north()`	`not_facing_north()`	Is Karel facing north?
`facing_east()`	`not_facing_east()`	Is Karel facing east?
`facing_south()`	`not_facing_south()`	Is Karel facing south?
`facing_west()`	`not_facing_west()`	Is Karel facing west?

Check out "Karel Reader", Chapter 10 for full reference

Which Type of Loop Should You Use?

`for` Loop

You know exactly how many times to repeat (definite loop)

vs.

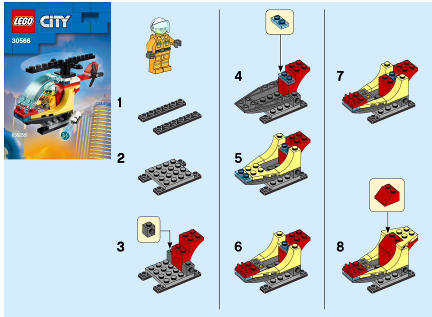
`while` Loop

You don’t know exactly how many times to repeat (indefinite loop)

`if-else` Statement

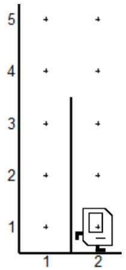
```
if condition:
    statements                # note indenting
else:
    statements


# Example:
def invert_beepers():
    if beepers_present():
        pick_beeper()         # note indenting
    else:
        put_beeper()          # note indenting
```

Decomposition



Break down a problem into more manageable sub-problems

- "Divide and conquer" approach
- Which sorts of tasks should be decomposed?
- Rule of Thumb: Each function should execute a single purpose.
- Code becomes easier to read!   For both you and everyone else.




```
turn_left()
while right_is_blocked():
        move()
turn_right()
move()
turn_right()
move_to_wall()
turn_left()
```

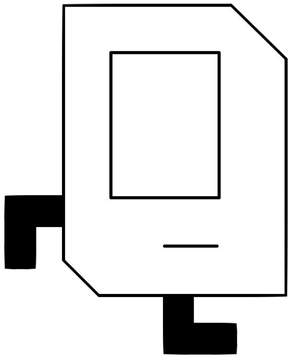
```
ascend_hurdle()
move()
descend_hurdle()
```

Navigating Karel:
Compass Direction vs. Relative Direction

Compass Direction

North

West



East

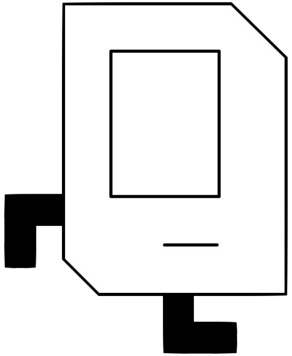
South

Navigating Karel:
Compass Direction vs. Relative Direction

Compass Direction | **VS.** | **Relative Direction**

North

West

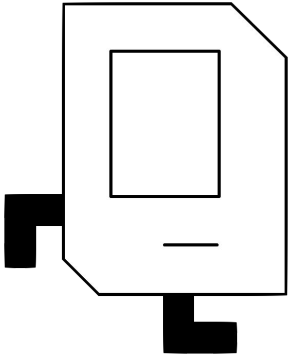


East

South

Left

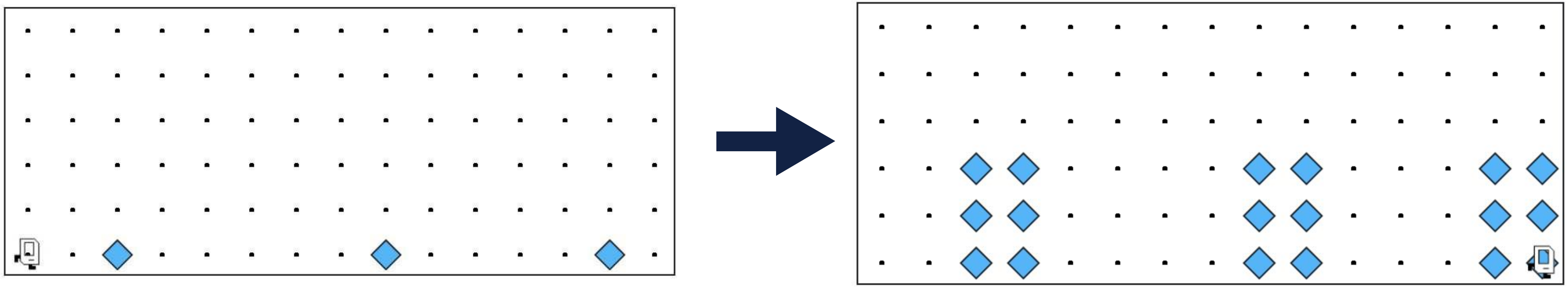
Back



Front

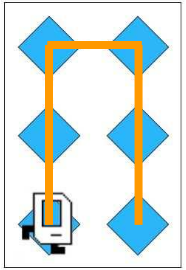
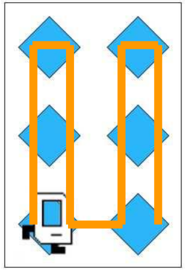
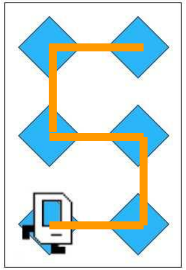
Right

Section Exercise: “Hospital Karel”



Section Exercise: “Hospital Karel”

Which option would you pick to build the 2x3 hospital? Pros vs Cons?

What if the hospital was another shape?